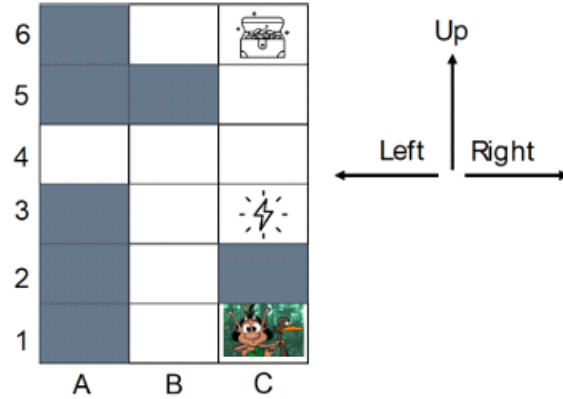


Q1. (1.5pts). Hugo is riding an ostrich to find the treasure. He has the map in hand, which is shown below. The white squares are accessible while the black ones are not. For each step, Hugo can only **move to the next accessible square** in one of the three directions—**Up, Left, and Right**. Each step consumes one unit of energy.



Assume that Hugo **starts at square C1** and the **treasure is at square C6**. The ostrich cannot move to the treasure without gaining some energy units along the way. Thus, Hugo needs to **first arrive at the energy source** (at square C3), and **then heads to the treasure**.

a) (0.5pt) Compute the Manhattan distance from accessible squares to the **energy source** and **treasure**.

Squares	B1	C1	B2	B3	C3	A4	B4	C4	C5	B6	C6
Power source											
Treasure											

b) (1pt) Draw a search strategy to achieve the goals specified in a), using A* with Manhattan distance. Since there are two goals, your strategy should be represented by one or several search trees, depending on how you define the heuristic functions with Manhattan distance. For each node on the search tree, specify a triple of (f, g, h)—g is the path cost, h is the heuristic value, and $f = g + h$ —and mark the best path found.

Q3. (2pts). Consider the following training set. For each calculation, write down the formula (with terms replaced by specific numbers) and the result. Numbers are rounded to three decimal places.

X_1	X_2	Target
2	-2	0
-1.5	-2	1
0.5	-1	0
-2	1	0

a) (1pt) Calculate the results of the **Forward pass** and the **Loss values** (no backpropagation required) for the following neural network.

$$W_1 = \begin{bmatrix} 1 & 2 \\ -2 & 1 \\ 1 & -2 \\ 2 & 2 \end{bmatrix}; B_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \end{bmatrix}; W_2 = [1 \quad -2 \quad 4 \quad 2]; B_2 = [1]$$

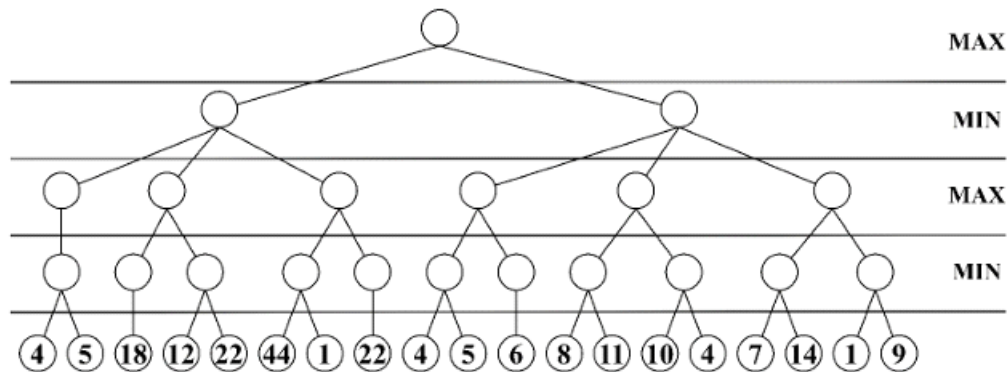
- Loss function** = $|\text{Target} - \text{prediction}|$
- Activation function**: $\text{step}(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$

b) (1pt) Consider a **perceptron** whose configuration is as follows.

- Weights = [0.4, -0.7] Bias = 0.1 (i.e., threshold $\theta = -0.1$) Learning rate = 0.5
 - Loss function = |Target - prediction| • Activation function: $\text{sigmoid}(x) = \frac{1}{1+e^{-x}}$
- Show how weights change in the first epoch.

Q5. (1pt). Consider the two-player game tree shown below. Nodes are evaluated in left-to-right order.

a) (0.5pt) Apply the **minimax algorithm** and write values directly for each node on the game tree.



b) (0.5pt) If there is any node pruned by using **alpha-beta pruning**, explain for each of these nodes.